

Differential Calculus

*Home work sheet # 6

Leibnitz's Theorem

1. If $y = \tan^{-1} x$, then show that $(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$.
2. If $y = \cot^{-1} x$, then show that $(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$.
3. If $y\sqrt{1-x^2} = \sin^{-1} x$, then show that $(1-x^2)y_{n+1} - (2n+1)xy_n - n^2y_{n-1} = 0$.
4. If $y = e^{\tan^{-1} x}$, then show that $(1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + n(n+1)y_n = 0$.
5. If $y = e^{m\sin^{-1} x}$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.
6. If $y = (\sin^{-1} x)^2$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.
7. If $\log_e y = a \sin^{-1} x$ then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$.
8. If $y = e^{m\cos^{-1} x}$ then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.
9. If $\log_e y = \tan^{-1} x$ then show that $(1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + n(n+1)y_n = 0$.
10. If $y = (\cos^{-1} x)^2$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.
11. If $\ln y = m\cos^{-1} x$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.
12. If $x = \tan(\ln y)$, then show that $(1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + n(n+1)y_n = 0$.

Indeterminate Forms

Find the limit using L' Hospital's rule:

1. $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$, 2. $\lim_{x \rightarrow 3} \frac{x-3}{3x^2-13x+12}$, 3. $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{\cos 3x}{x^2} \right)$, 4. $\lim_{x \rightarrow \pi} \frac{\sin x}{x-\pi}$
5. $\lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3}$, 6. $\lim_{x \rightarrow +\infty} \frac{e^{3x}}{x^2}$, 7. $\lim_{x \rightarrow 0} \frac{a^x - 1 - x \log a}{x^2}$, 8. $\lim_{x \rightarrow 0} (e^x + x)^{\frac{1}{x}}$
9. $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$, 10. $\lim_{x \rightarrow \pi} (x-\pi) \cot x$, 11. $\lim_{x \rightarrow 0} \frac{\ln(\sin x)}{\ln(\tan x)}$, 12. $\lim_{x \rightarrow \infty} x e^{-x}$
13. $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$, 14. $\lim_{x \rightarrow 0} \frac{\sin x}{x^2}$, 15. $\lim_{x \rightarrow \infty} \frac{x}{e^x}$, 16. $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x e^x} \right)$, 17. $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 5x}$.

*These problems are for the students only as home work. Search the reference books for more examples.